

Liam Meagher

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EDUCATION

University of Southern California, M.S. in Mechanical Engineering **Expected May 2028**

University of California, Los Angeles, B.S. in Computer Science & Engineering **September 2021 – June 2025**

EXPERIENCE

Chassis Lead Engineer, Flagship Combat Robotics – Los Angeles, CA **January 2023 – June 2025**

- Spearheaded the mechanical design and fabrication of several autonomous vertical/horizontal spinner weapon systems
- Applied **GD&T** and **tolerance stack-up analysis** to ensure proper fits, alignment, and **DFM-ready component designs**
- Leveraged **SolidWorks CAD + FEA** to validate chassis and weapon subassemblies under **high-shock loading (>11 kN)**
- Employed **Ansys Discovery topological optimization** to reduce mass by **42% (10.3 lb)** while maintaining stiffness targets
- Produced **fabrication drawings** and **release packages** with datums, tolerances, and inspection-critical dimensions
- Manufactured critical **AR500 steel** and **7075-T6 aluminum** components utilizing both manual and CNC milling operations
- Accelerated chassis and drivetrain prototyping cycles by **30%** through rapid iteration with 3D printing and laser cutting
- Instructed **30+ team members** on researching, designing, and machining **4 competition-ready combat robots**

Autonomous Systems Software Developer, Bruin Underwater Robotics – Los Angeles, CA **September 2023 – June 2024**

- Developed autonomy control software for an underwater robot, implementing real-time **feedback control loops** for depth
- Improved closed-loop stability by **reducing oscillations by 40%** through **PID tuning** and **dynamic response testing**
- Generated **2000+ simulated sensor frames** and produced **5+ hours of synthetic mission data** using **Unity's simulation** tools
- Conducted simulation-driven reliability testing across **8 mission scenarios**, increasing autonomous control stability by **45%**
- Integrated control software with thrusters, depth sensors, and embedded hardware to ensure robust real-world operations

Programming & Game Design Lead, ACM Studio – Los Angeles, CA **September 2021 – January 2024**

- Developed and implemented rotating classes, structures, and colliders for **4 weapon systems** and **16 unique animations**
- Collaborated on **rapid prototyping**, simulation-based **general debugging**, and **asset integration** under tight time constraints
- Implemented AI-driven decision systems with **4 scaling difficulty levels**, modeling adaptive opponent behavior and skills
- Optimized Unity's performance and reduced memory usage by **28%** to support **large-scale simulation testing** of opponents

Full-Stack Web Developer, UCLA Marketplace – Los Angeles, CA **January 2022 – January 2023**

- Designed and developed a peer-to-peer platform enabling **200+ UCLA students** to buy, sell, and trade items without fees
- Implemented secure SQL-based **authentication** and **transaction infrastructure**, ensuring reliable data integrity across users
- Maintained team codebase through structured **Git workflows**, resolving integration issues and supporting consistent code
- Achieved **98% uptime** with **Node.js landing page** and secure **peer-to-peer messaging** board across student accounts

Activities & Outdoor Programming Director, Scouting America – Avalon, CA **May 2021 – August 2022**

- Operated a weekly outdoor activity program supporting over **4,000 participants** with a budget of over **\$390,000**
- Directed, evaluated, and appraised **30 staff members** and provided constructive feedback to develop their skillsets
- Maintained a varied inventory of program equipment to support and grow staff morale of **170+ full-time staff**
- Created a thorough end-of-year report for future directors to track and monitor responsibilities between seasons
- Recruited, interviewed, hired, and trained relevant staff members for subsequent 2022 and 2023 summers

SKILLS

Software: SolidWorks (CAD/FEA), Onshape (CAD/PDM), Ansys Discovery, MATLAB, Linux, Excel, Unity, Unreal Engine

Manufacturing: CNC (Mill/Lathe), 3D Printing (FDM), G-Code, Soldering, Drill Press, GD&T, Design for Manufacturability (DFM)

Languages: C, C++, C#, Haskell, Java, JavaScript, Python, R, RISC-V Assembly, SQL, Verilog

Libraries & Tools: NumPy, Pandas, Matplotlib, OpenCV, Scikit-Learn, PyTorch, React, Node.js, Docker, Git, Arduino, ROS, ROS 2